Chapter Two

Introduction to Data Science

Topics Covered

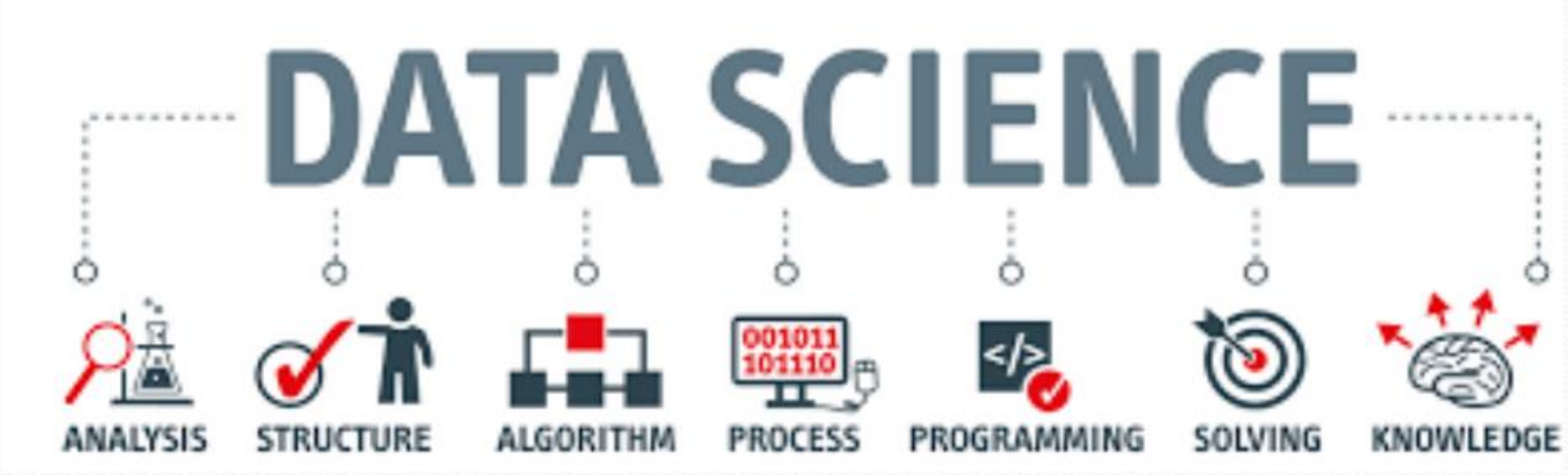
- An Overview of Data Science
- Data and information
- Data types and representation
- Data Processing Cycle
- Data Value Chain (Acquisition, Analysis ,Curating, Storage, Usage)
- Basic concepts of Big data

2.1 Overview of Data Science

- **What is Data science?**
  - ✓ A multi-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured, semi structured and unstructured data.
  - ✓ is much more than simply analyzing data.
- Examples of data
  - Your notebook
  - Prices of items in supermarket
  - Files in computer
  - Barcode etc.

Overview…

- Data science continues to evolve as one of the most promising and in-demand career paths for skilled professionals
- To be a successful data professional in today's market requires to advance past traditional skills of analyzing large amounts of data by data mining and programming skills.



Cont. . . Data Science Experts/Scientist?

- ✓ **Data scientists** are analytical experts who utilize their skills in both technology and social **science** to find trends and manage **data**.
- ✓ They use industry knowledge, contextual understanding, uncertainty of existing assumptions to uncover solutions to business challenges..

- ✓ Need a strong quantitative background in statistics and linear algebra as well as programming knowledge
- ✓ Need to be curious and result-oriented
- ✓ Must master the full spectrum of the **data science life cycle** and possess a level of flexibility and understanding to maximize returns.

2.2 Data and Information

- **Data?**
  - ✓ A representation of facts, concepts, or instructions in a formalized manner, which should be suitable for communication, interpretation, or processing by human or electronic machine.
- **Information?**
  - ✓ Organized or classified data, which has some meaningful values for the receiver.
  - ✓ A processed data on which decisions and actions are based.
  - ✓ Principle of information - processed data must qualify for the following
    - Timely – Information should be available when required.
    - Accuracy – Information should be accurate.
    - Completeness – Information should be complete.

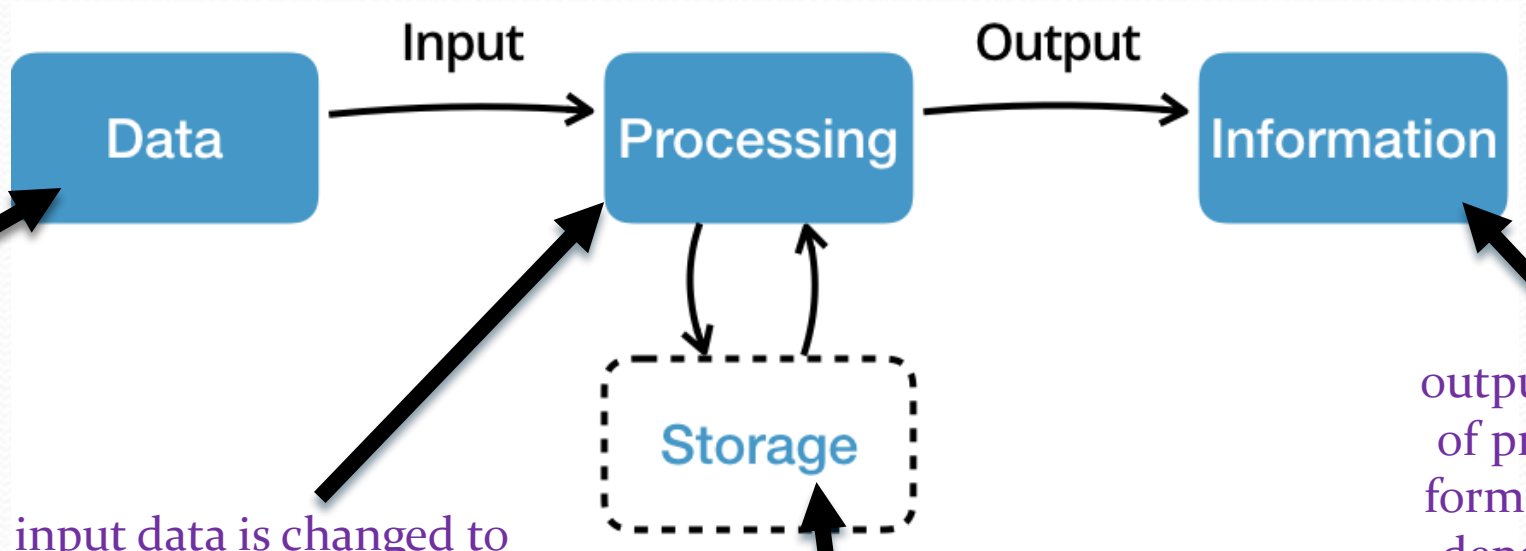
Summery: Data Vs. Information

Data	Information
Described as unprocessed or raw facts and figures	Described as processed data
Cannot help in decision making	Can help in decision making
Raw material that can be organized, structured, and interpreted to create useful information systems.	Interpreted data; created from organized, structured, and processed data in a particular context.
‘groups of non-random’ symbols in the form of text, images, and voice representing quantities, action and objects'.	Processed data in the form of text, images, and voice representing quantities, action and objects'.

2.3 Data Processing Cycle

- Re-structuring or re-ordering of data by people or machine to increase their usefulness and add values for a particular purpose.
- The set of operations used to transform **data** into useful **information.**

Data Processing Cycle



input data is prepared in some convenient form for processing e.g. electronic computers

input data is changed to produce data in a more useful form e.g. calculating CGPA

Produced information need to be stored for future usage

output data is the result of processing step and form of the output data depends on the use of the data.

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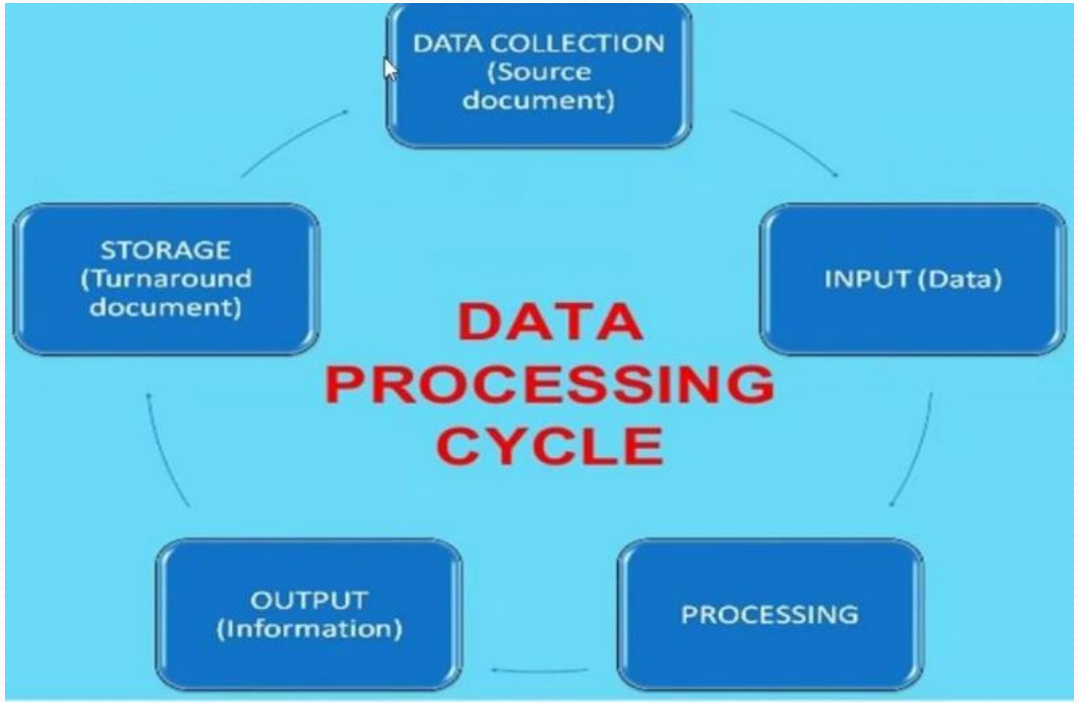

Input Stage	Processing Stage	Output Stage	Storage Stage
Data Collection	Performing instructions	Decoding	Storing data
Data Capture	Transform raw data into information	Presenting data to user	Retrieve data
Encoding			
Data Transmission			
Data communications			

- 35,34,33,32,36
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday

- Arranging
- Sorting
- Combining
- Mathematical operations

- Temperature
- Monday 35 °C
- Tuesday 34 °C
- Wednesday 33 °C
- Thursday 32 °C
- Friday 36 °C

2.4 Data types and their representation

- Data type defines the operations that can be done on the data, the meaning of the data, and the way values of that type can be stored.
- Data types can be described from diverse perspective

(a) Computer programming perspective

- ✓ Integers (int) --- whole numbers
- ✓ Booleans (bool) -- restricted to one of two values: true or false
- ✓ Characters (char) -- store a single character (symbol)
- ✓ floating-point numbers (float) --- store real numbers
- ✓ alphanumeric strings (string) --- group of characters

(b) Analytics perspective

- ✓ Structured Data:
- ✓ Semi-structured Data:
- ✓ Unstructured Data

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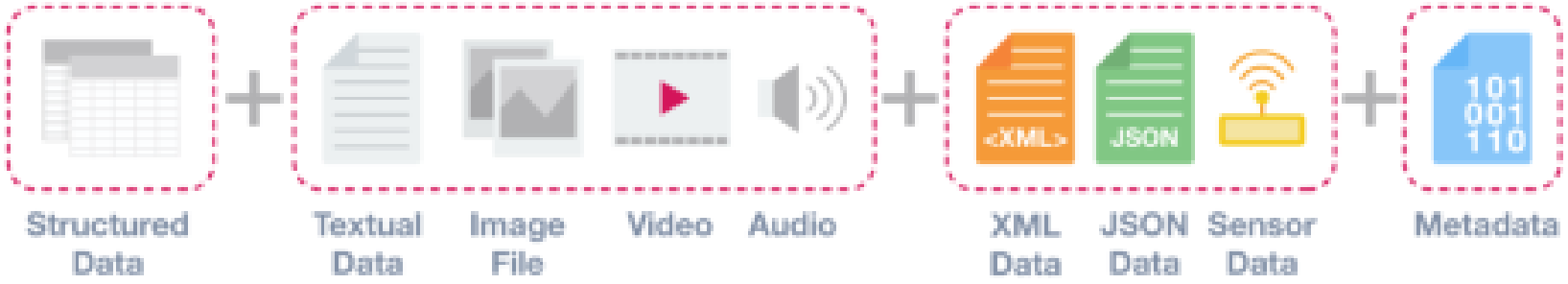


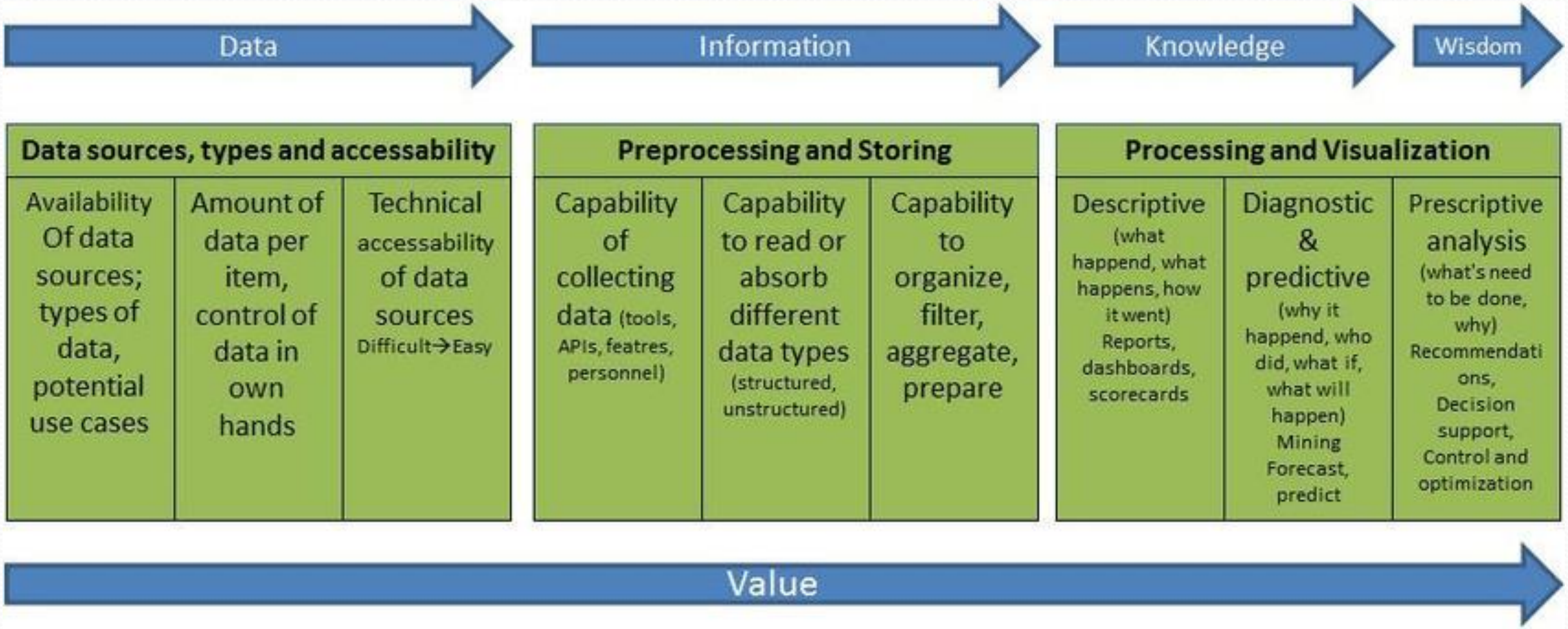
- Structured Data:
  - ✓ adheres to a pre-defined data model and straightforward to analyze
  - ✓ take a tabular format. E.g. Excel files or SQL databases
- Semi-structured Data:
  - ✓ does not conform with the formal structure of data model.
  - ✓ Contains tags or other markers for separation semantic elements enforce hierarchies of records and fields within the data
  - ✓ Fore example: JSON and XML
- Unstructured Data
  - does not have a predefined data model or is not organized in a pre-defined manner.
  - Typically text-heavy but may contain data such as dates, numbers, and facts as well.
  - Examples: audio, video files or No-SQL databases.

✓ Metadata - data about data that provides additional information about a specific set of data.

e.g. photographs metadata - describe when and where the photos were taken.

2.5 Data value chain

- Introduced to describe the information flow within a big data system as a series of steps needed to generate value and useful insights from data.



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- ensuring data trustworthiness, accessibility, reusability
- content creation, selection, classification, transformation, validation, and preservation

- Ensuring the needs of fast access to the data
- RDBMS & NoSQL
- ACID (Atomicity, Consistency, Isolation, & Durability)

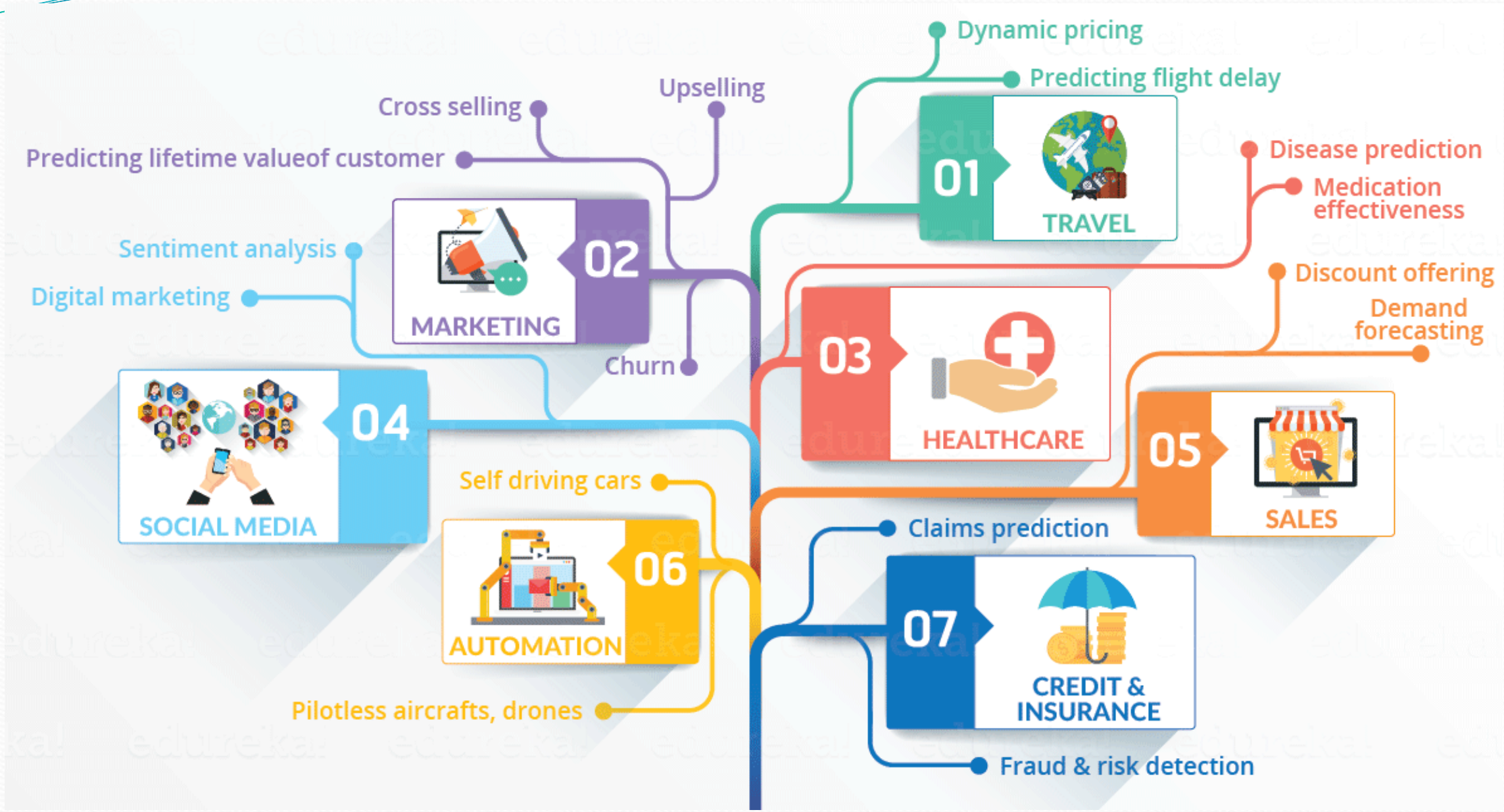
- Deliver in low latency,
- Predictable in both capturing data and executing queries
- Be able to handle very high transaction data volumes,
- Flexible and dynamic in a distributed environment

Data Acquisition	Data Analysis	Data Curation	Data Storage	Data Usage
• Structured data • Unstructured data • Event processing • Sensor networks • Protocols • Real-time • Data streams • Multimodality	• Stream mining • Semantic analysis • Machine learning • Information extraction • Linked Data • Data discovery • 'Whole world' semantics • Ecosystems • Community data analysis • Cross-sectorial data analysis	• Data Quality • Trust / Provenance • Annotation • Data validation • Human-Data Interaction • Top-down/Bottom-up • Community / Crowd • Human Computation • Curation at scale • Incentivisation • Automation • Interoperability	• In-Memory DBs • NoSQL DBs • NewSQL DBs • Cloud storage • Query Interfaces • Scalability and Performance • Data Models • Consistency, Availability, Partition-tolerance • Security and Privacy • Standardization	• Decision support • Prediction • In-use analytics • Simulation • Exploration • Visualisation • Modeling • Control • Domain-specific usage
Data acquisition is the process of gathering, filtering and cleaning data, before any data analysis can be carried out.	Data analysis is concerned with making the raw data acquired amenable to use in decision-making as well as domain-specific usage.	Data curation is the active management of data over its life cycle to ensure it meets the necessary data quality requirements for its effective usage.	Data storage is the persistence and management of data in a scalable way that satisfies the needs of applications.	Data usage covers the data-driven business activities that need access to data, its analysis, and the tools needed to integrate the data analysis within the business activity

- Exploring, transforming, and modelling data with the goal of highlighting relevant data
- Synthesizing and extracting useful hidden information with high potential from a business point of view

- **Infrastructure**

Cont. . . Use case of Data Science



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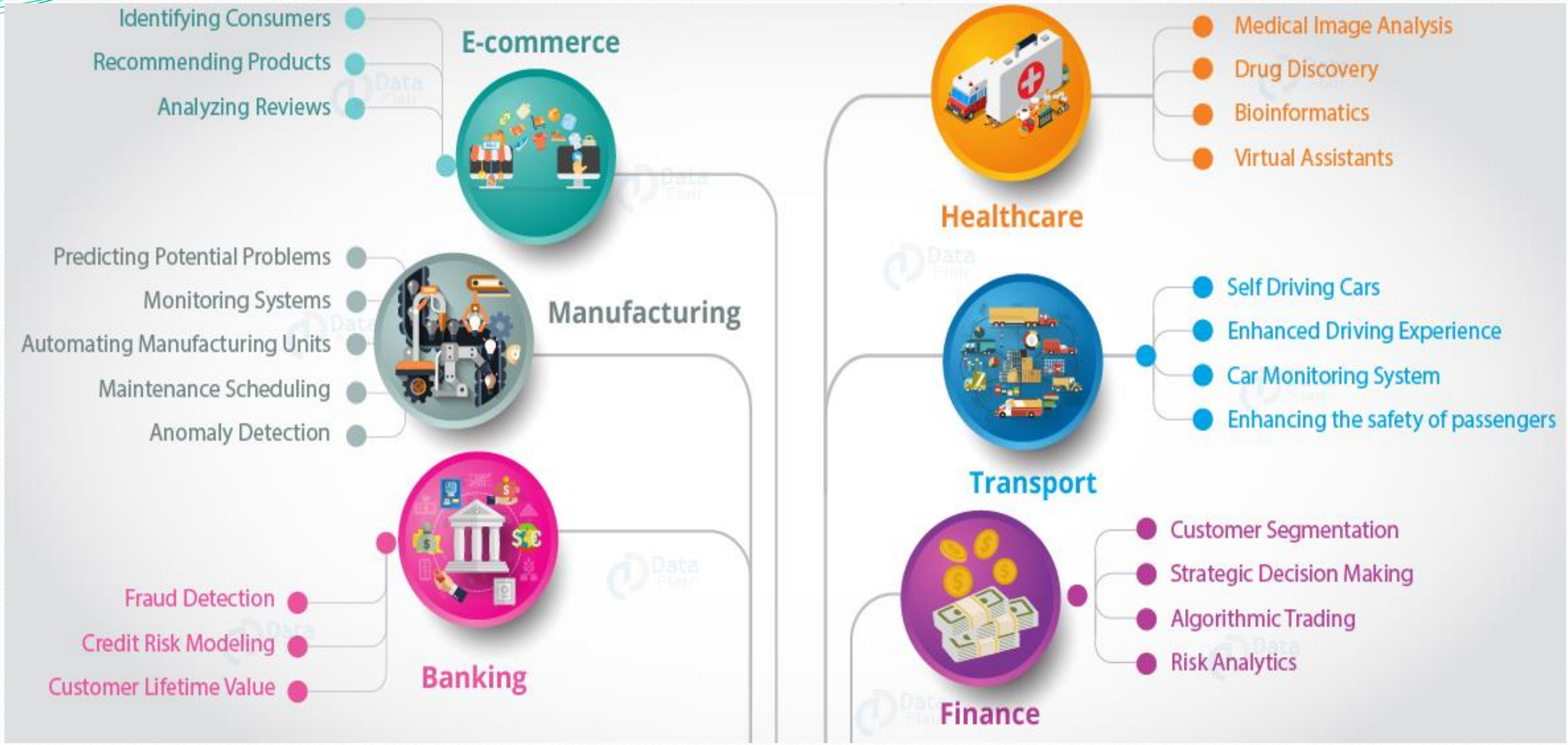
Application domain of Data Science



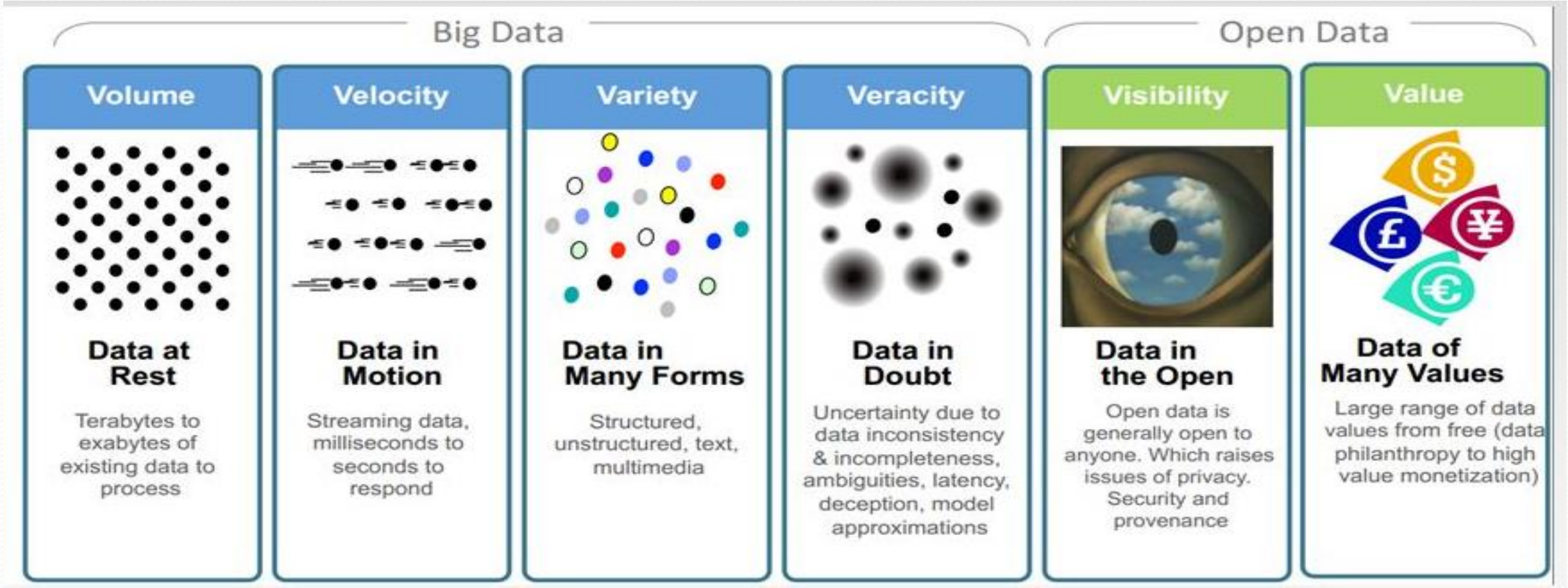
2.6 Basic concepts of Big data

- **Big data** is a blanket term for the non-traditional strategies and technologies needed to gather, organize and process insights from large datasets.

- Big data refers to:
  - ✓ large and complex datasets that it becomes difficult to process using on-hand database management tools or traditional data processing applications
  - ✓ The category of computing strategies and technologies that are used to handle large datasets

- Goal of Big data:
  - ✓ To surface insights and connections from large volumes of heterogeneous data that would not be possible using conventional methods

Cont. . . Characteristics of Big data



- The amount of data from myriad source
- large amounts of data Zeta bytes (Massive datasets)

- The speed at which data are generated
- Data is live streaming or in motion
- Realtime

- The types of data
- Data comes in many different forms from diverse sources

- Data trustworthiness (the degree to which big data can be trusted)
- Data accuracy
  How accurate is it?

- The way in which the big data can be used and formatted
- To whom the data are accessible?

- Business value of the data collected
- Uses and purpose of data

Hadoop and its Ecosystem

- Hadoop is an open-source framework intended to make interaction with big data easier.
- It is inspired by a technical document published by Google.
- It allows for the distributed processing of large datasets across clusters of computers using simple programming models.

- The four key characteristics of Hadoop

✓ **Economical**: Its systems are highly economical as ordinary computers can be used for data processing.

✓ **Reliable**: It is reliable as it stores copies of the data on different machines and is resistant to hardware failure.

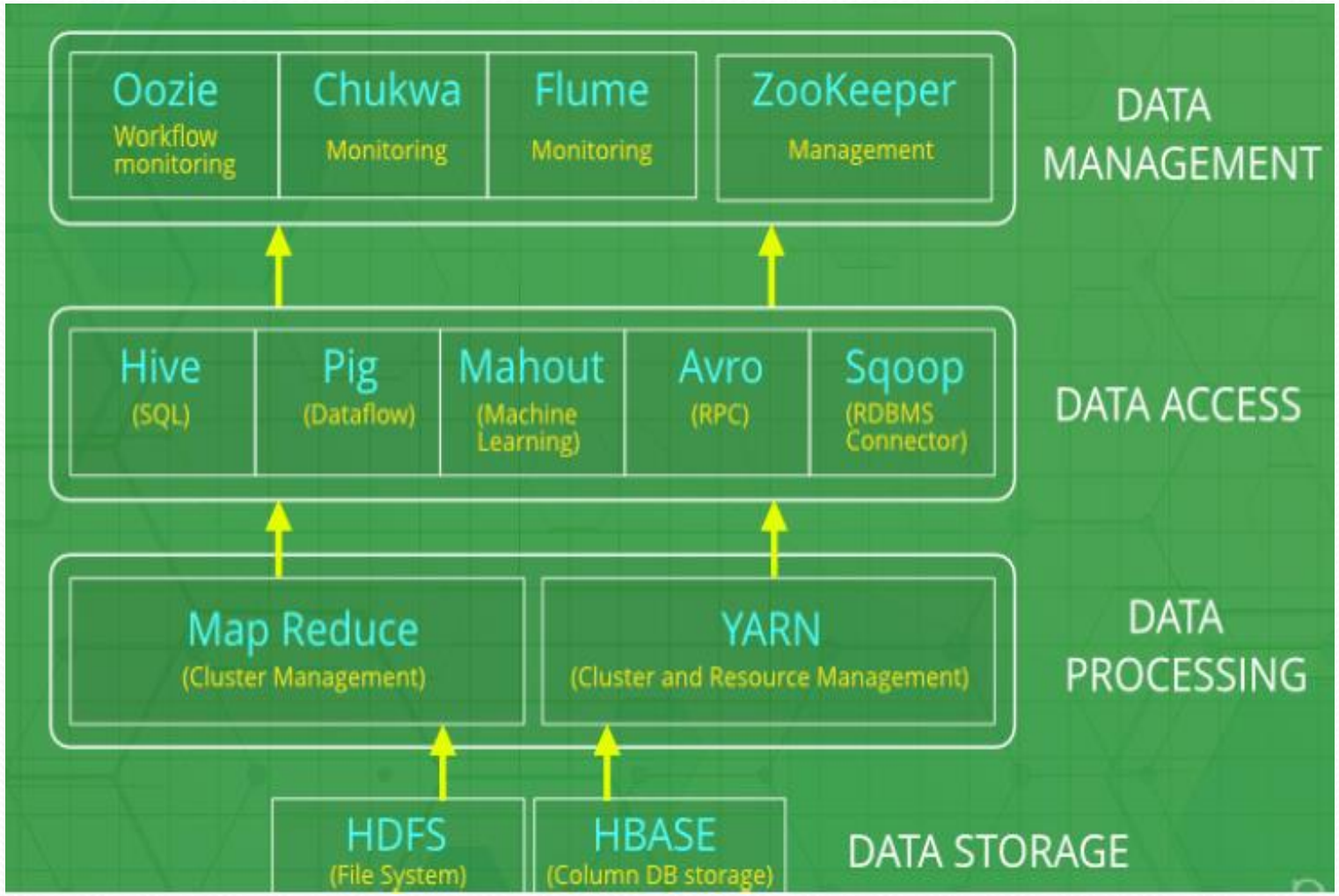
✓ **Scalable**: It is easily scalable both, horizontally and vertically. A few extra nodes help in scaling up the framework.

✓ **Flexible**: It is flexible and you can store as much structured and unstructured data as you need to and decide to use them later.

Cont. . .

The 4 core components of Hadoop and its Ecosystem

- The 4 core components of Hadoop includes
  - ✓ Data Management,
  - ✓ Data Access,
  - ✓ Data Processing
  - ✓ Data Storage.



The Hadoop Ecosystem

Cont. . .

The 4 core components of Hadoop and its Ecosystem

- Hadoop ecosystem comprises of the following components
  - ✓ **HDFS**: Hadoop Distributed File System
  - ✓ **YARN**: Yet Another Resource Negotiator
  - ✓ **MapReduce**: Programming based Data Processing
  - ✓ **Spark**: In-Memory data processing
  - ✓ **PIG, HIVE**: Query-based processing of data services
  - ✓ **HBase**: NoSQL Database
  - ✓ **Mahout**, **Spark MLLib**: Machine Learning algorithm libraries
  - ✓ **Solar, Lucene**: Searching and Indexing
  - ✓ **Zookeeper**: Managing cluster
  - ✓ **Oozie**: Job Scheduling

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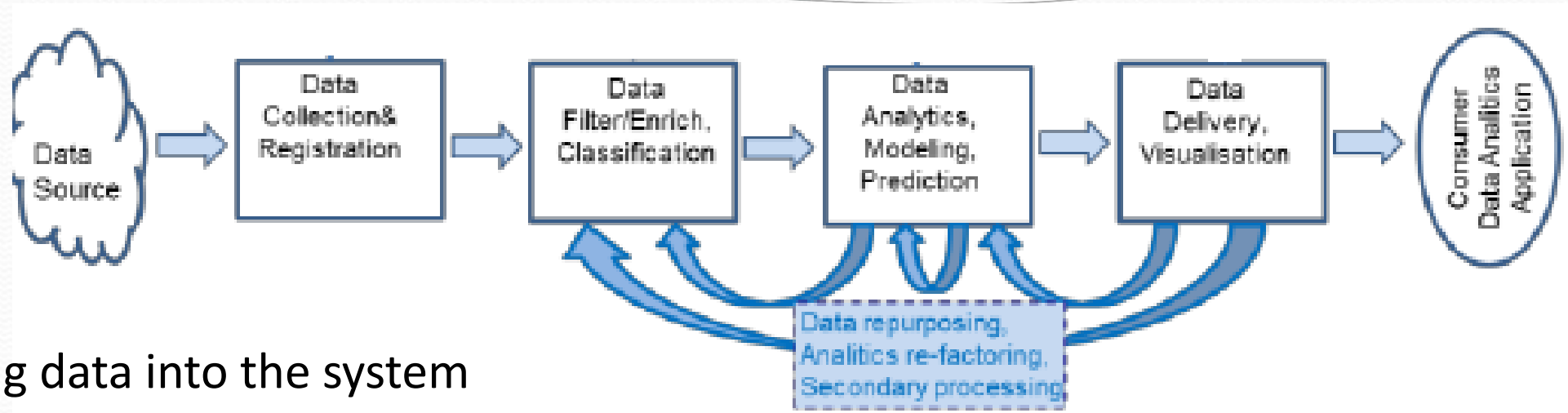
The Big data life cycle with Hadoop



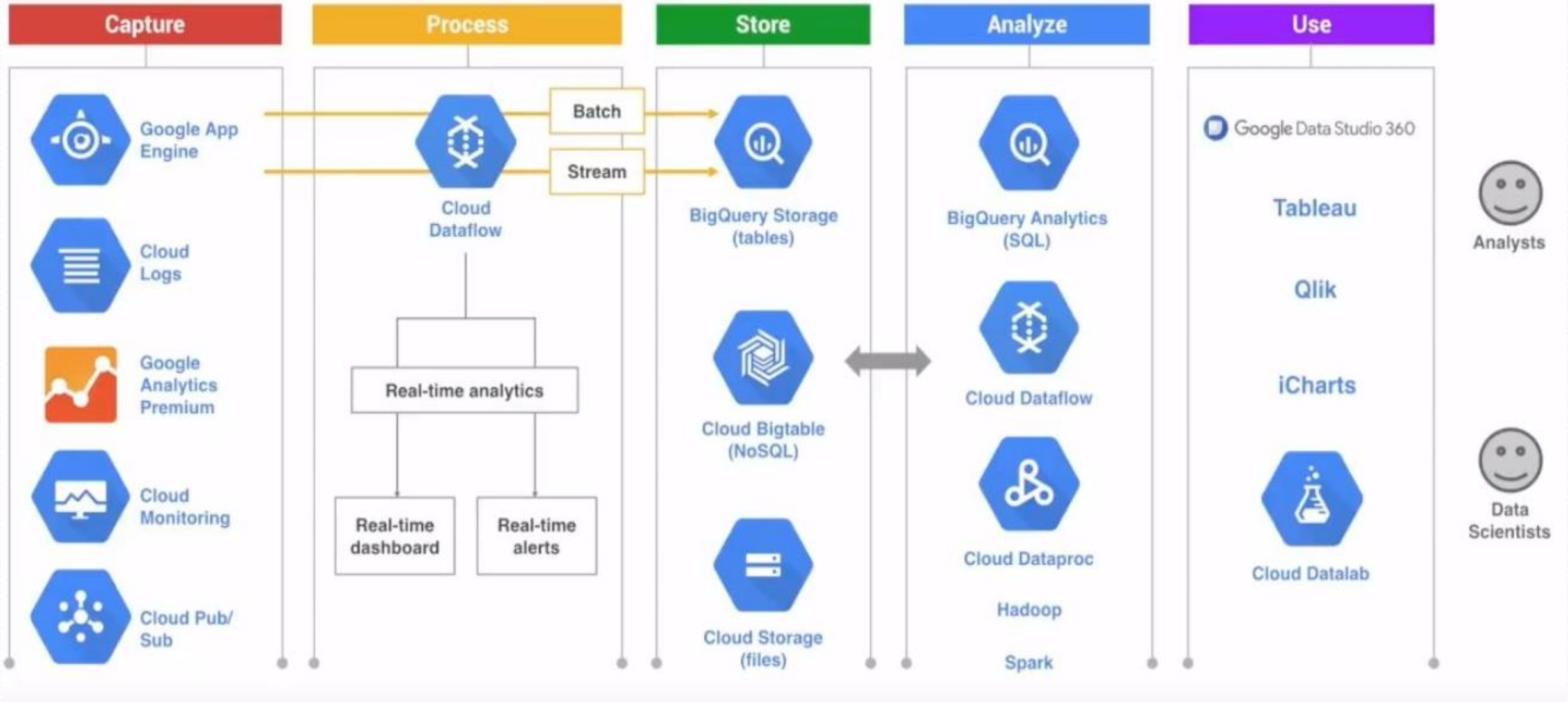
- **Stage 1-** Ingesting data into the system
  - ✓ The data is ingested or transferred to Hadoop from various sources such as relational databases, systems, or local files.
- **Stage 2-** Processing the data in storage (stored and processed)
  - ✓ The data is stored in the distributed file system, HDFS, and the NoSQL distributed data, HBase. Spark and MapReduce perform data processing.
- **Stage 3-** Computing and analyzing data
  - ✓ by processing frameworks such as Pig, Hive, and Impala.
- **Stage 4-** Visualizing the results (Access)
  - ✓ by tools such as Hue and Cloudera Search

Want to know more about Hadoop?

- Hadoop in 5 minutes:
  - https://www.youtube.com/watch?v=aReuLtYoYMI
- Introduction to Hadoop:
  - https://www.youtube.com/watch?v=iANBytZ26MI

Cont. . .

Big data lifecycle at Google Cloud platform



End of Chapter 2
(Data Science)

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Review Questions

1. Discuss the difference of Big data and Data Science.
2. Briefly discuss the Big data life cycle.
3. List and explain Big data application domains with example.
4. What is Clustered Computing? Explain its advantages.

Thank you!